

6. Build and deploy the model using the  [Monitor & Tune](#) action button under the Hardware Tab of your model.
7. In the section labelled [Position Select](#) enter the suggested values given for vectors [A](#), [B](#), and [C](#). Note that the constant beta (β) has also been defined for you and you can use 'beta' directly.
8. Set [Select Position](#) to 1. This will place the end-effector directly above the base without the manipulator being fully stretched out. If your earlier implementation of Equations 3 and 6 were correct you should observe a value close to zero shown in the display labelled [Condition 1](#) and a non-zero value in the display labelled [Condition 2](#). Otherwise, re-examine your Simulink code.
9. Set [Select Position](#) to 2 to fully stretch out the manipulator without the end-effector directly being above the base (the shoulder angle can be any value not close to 0). If your implementation of the Singularity Conditions 1 and 2 were correct, the display labelled Condition 2 should show a value close to 0 and Condition 1 should be non-zero.
10. Set [Select Position](#) to 3 to place the end-effector above the base with the arm fully stretched out. Verify that in this case, the values shown in [Condition 1](#) and [Condition 2](#) both show values close to 0 for the corresponding singular conditions.
11. Stop the model.

Mathematical Approach

In this approach you will examine the Jacobian and its invertibility to identify any joint configurations which will result in a singularity.

1. Uncomment the section labelled [#2 - Identification via Jacobian and its inverse](#).
2. Build and deploy the model using the  [Monitor & Tune](#) action button under the Hardware Tab of your model.
3. Set [Select Position](#) to 1. This will place the end-effector directly above the base without the manipulator being fully stretched out.
4. While the manipulator is in this position, examine the values and record the approximate values shown in the displays labelled [Jacobian J](#), [Jacobian Inverse](#), and [Condition Number](#). Take notes on the column vectors that make up the Jacobian. Does the condition number relate to any element of the Jacobian's inverse?
5. Repeat the previous step for the remaining two positions (i.e. positions B and C). Make a note of your findings.
6. Stop the model.

Visual Approach

In this approach you will place the manipulator in a series of singularity positions. For each position, the goal is to observe if the end-effector loses movement along any of the task space X, Y, and Z axes when the manipulator's joints are actuated.

1. Uncomment the section labeled [#3 - Identification via Task and Joint Velocities](#) in the Simulink model.

- Complete the section such that the scopes labelled **Task Velocities Expressed in Frame 0** and **Task Velocities Expressed in Frame 4** display the X, Y, Z components of the end-effector's velocity vector expressed in Frame 0 and Frame 4, respectively. *Hint: you can use the rotation matrix R_{04} to change the expression of a vector from frame 4 to frame 0, and the inverse of a rotation matrix is its transpose.*
- Build and deploy the model using the  **Monitor & Tune** action button under the Hardware Tab of your model. Set **Select Position** to 1. This will place the end-effector directly above the base without the manipulator being fully stretched out.
- Open the scopes labelled **Task Velocities Expressed in Frame 0** and **Task Velocities Expressed in Frame 4**. With the arm in position A, use the manual switch placed before the Disturbance subsystem to move the joints one by one. The scopes will display the applied velocity disturbances, as well as the X, Y, Z components of the end-effector's velocity when expressed in frames 0 and 4. Sample result shown in Figure 2 indicates that when all four of the joints have been actuated, the end-effector only moves along the X (red) and Z (green) axes when expressed in frame 0. In other words, movement along the Y axis is lost. Use Table 1 to record your results.
- Repeat the previous step for the remaining two positions, i.e. setting **Select Position** to 2 and 3. Record your results in Table 1.
- Stop the model. While supporting the arm, turn OFF the power using the switch at the rear end of the base and move the arm to the rest position.

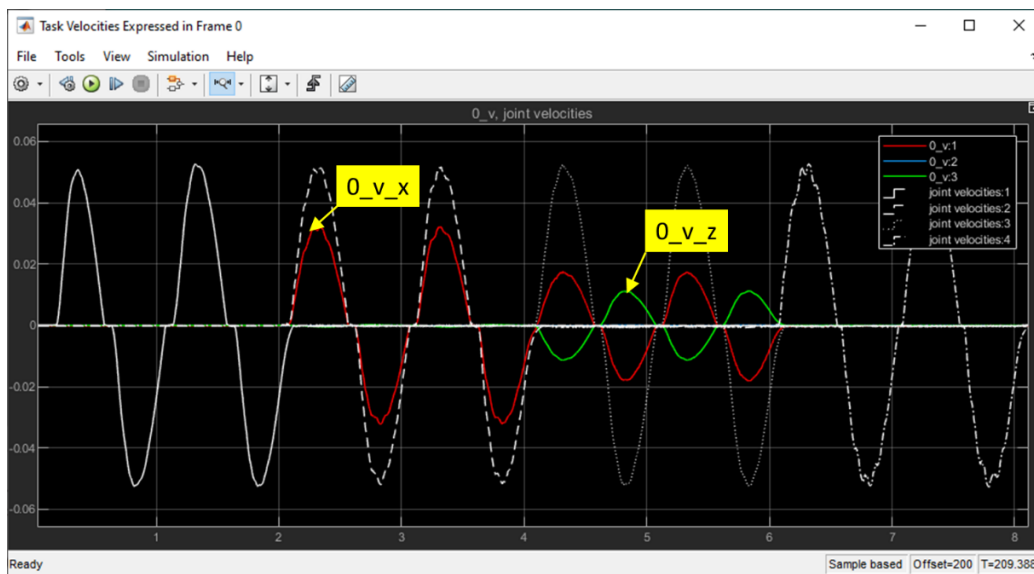


Figure 2 – Sample result for position A showing loss of velocity along the Y axis when expressed in frame 0

	Lost velocity components	
	Frame 0	Frame 4
Position A	0_v_y	
Position B		
Position C		
Position D		

Table 1 – Recorded loss of movement along X, Y, Z axes expressed in frames 0 and 4